Learn Beneficial Noise as Graph Augmentation

PiNGDA: Positive-incentive Noise driven Graph Data Augmentation

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Motivation: Why Existing Graph Augmentations Fall Short

The Problem

- Graph topology is complex (non-Euclidean)
- Random edge/node dropping may disrupt critical structure
- Heuristic methods (GCA, DropEdge) cause instability
- No principled way to define 'safe' perturbations
- Unstable augmentations degrade GCL pre-training

Key Insight: π-Noise

Noise ε is beneficial if it reduces task complexity:

$$I(T, E) > 0 \iff H(T) > H(T \mid E)$$

- Noise must reduce task entropy H(T)
- Standard GCL = point estimate of π-noise
- Heuristic augmentations ≈ poor point estimates
- Solution: learn the noise distribution

Theoretical Framework: Bridging π-Noise and GCL

1 Task Entropy

Introduce Gaussian auxiliary variable α. Task entropy:
$H(T) = E[H(N(0, f(\ell(u;\theta))))]$

2 π-Noise Objective

Maximize mutual information:
$\max_{E,\theta} I(T,E) \iff \max_{E,\theta} -H(T|E)$

3 GCL Connection

Predefined augmentation ≡ Dirac delta $p(\varepsilon|u) \to \delta_{\varepsilon_0}(\varepsilon)$
Reduces to InfoNCE — standard GCL is a point estimate

4 PiNGDA Solution

Learn $p_\psi(\varepsilon|u)$ with a trainable noise generator:
$L_\pi = -(1/n) \sum_u E_{\varepsilon\sim p_\psi} [\log \kappa_\theta(u,\varepsilon)]$

PiNGDA Framework Architecture

π-Noise Generator

Topological Noise (ε_{edge})

MLP on node pairs (u, v)

$p(\varepsilon_v|u) = \mathrm{Bern}(g_\psi(u,v))$

Gumbel-Softmax for gradients

Attribute Noise (ε_{attr})

$p(\varepsilon_{attr}|u) = N(\mu, \Sigma)$

MLP learns μ and σ per node

$\varepsilon_{attr} = \mu + \sigma \cdot \epsilon, \ \epsilon \sim N(0,I)$

→

Augmented Graph

Original G + topological & attribute noise → G_ε

→

Contrastive Learning

GNN Encoder ($G \rightarrow Z$)

GNN Encoder ($G_\varepsilon \rightarrow Z_\varepsilon$)

Loss: L_π (InfoNCE variant)

Results: Node Classification — 7 Benchmark Datasets

Method	Cora	CiteSeer	PubMed	Wiki-CS	Photo	Phy	Rank
GCA	85.20	71.76	87.07	81.26	93.19	94.96	3.4
BGRL	83.80	71.51	85.60	81.38	93.19	95.54	4.2
AD-GCL ✓	83.88	68.74	84.23	80.57	91.92	95.63	6.0
JOAO ✓	82.77	72.09	83.83	80.43	78.23	94.30	6.8
PiNGDA ✓	**86.25**	**72.44**	**87.34**	**82.07**	**93.29**	**95.81**	**1.4**

PiNGDA ranks #1 across all datasets with lowest variance — outperforming both heuristic and learnable baselines

Ablation Study: Augmentation Strategy Comparison

Cora

Edge × Feature →	Accuracy
No Aug. × No Aug.	82.59
No Aug. × Random	84.40
No Aug. × Learnable	84.28
Random × Random	85.84
Learnable × Learnable	**86.25**

PubMed

Edge × Feature →	Accuracy
No Aug. × No Aug.	85.11
No Aug. × Random	86.61
No Aug. × Learnable	86.81
Random × Random	86.89
Learnable × Learnable	**87.34**

Wiki-CS

Edge × Feature →	Accuracy
No Aug. × No Aug.	80.34
No Aug. × Random	81.05
Random × Random	81.61
Random × Learnable	82.03
Learnable × Learnable	**82.17**

Learnable (both edges + features) consistently achieves the best accuracy

What Does PiNGDA Learn? Visualization & Efficiency

Topological Noise Visualization

PiNGDA learns to:

- Preserve intra-class edges → keep class signal
- Remove inter-class edges → reduce noise
- Result: more informative augmented views

Efficiency Profile

Memory	Less memory than most GCL methods
Runtime	Comparable to existing approaches
Trade-off	Best accuracy-efficiency balance
Scale	Tested on ogbn-arxiv (169K nodes)
Plug-in	Compatible with GRACE, Sp2GCL, etc.

Conclusion

1 Theory

Standard GCL is equivalent to a point estimation of π-noise. PiNGDA proves this connection via information entropy and proposes a principled improvement.

2 Method

A trainable noise generator learns beneficial perturbations on both topology (Gumbel-Softmax Bernoulli) and node attributes (Gaussian reparameterization).

3 Results

PiNGDA achieves rank #1 across 7 node classification benchmarks with the lowest variance — more stable and accurate than all baselines.